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INTERIM FEDERAL SPECIFICATION

RODENTICIDE, ANTICOAGULANT, UNIVERSAL, CONCENTRATE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers one type of rodenticide concentrate suitable for the control of commensal rats and mice when mixed with a suitable bait.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

RR-S-366 - Sieve, Test.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

FSC 6840

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Laws and Regulations:

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.
49 CFR 170 - Department of Transportation Regulations for Hazardous Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date on invitation for bids or request for proposal shall apply.

The United States Pharmacopeia.

(The U.S. Pharmacopeia is available from the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

3. REQUIREMENTS

3.1 Material. The composition of the rodenticide shall be as specified in table I. The anticoagulant may be the sodium salt, calcium salt, or acid form of any of the optional anticoagulants specified in table I.

TABLE I. Composition

Ingredient	Percent		Test Method Paragraph
	Minimum	Maximum	
Anticoagulant options: 1/			
Fumarin	0.3	0.33	4.4.2.1
Pival	0.3	0.33	4.4.2.2
Diphacinone	0.05	0.055	4.4.2.3
Chlorophacinone	0.05	0.055	4.4.2.4
PMP	0.5	0.55	4.4.2.5
Tetrasodium ethylenediamine tetraacetate 2/	3.0	3.3	4.4.2.6
Sodium benzoate	3.0	3.3	4.4.2.7
Sugar 3/	Remainder		-------

1/ See 6.3.
2/ Not required in a fumarin formulation.
3/ Up to 5 percent of the rodenticide may be cornstarch.

3.2 Particle size. The rodenticide concentrate shall be a fine, free-flowing material devoid of lumps. No less than 95 percent of the material shall pass through a 60-mesh U.S. Standard Screen (dry test) when tested as specified in 4.4.2.8.

3.3 Moisture. The rodenticide concentrate shall not lose more than 1 percent of its weight when tested as specified in 4.4.2.9.

3.4 Performance. The toxicity and acceptability of the rodenticide concentrate shall be such that not more than 20 percent of the test animals shall survive when the product is tested as specified in 4.4.2.10.

3.5 Registration. The product shall be registered and labeled in compliance with 40 CFR 162, Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105.

4.3 Examination of preparation for delivery.

4.3.1 Sampling procedure. The lot size for this examination shall be defined as the number of shipping containers offered for inspection at one time. The sample unit shall be defined as one shipping container fully marked and prepared for delivery. The samples shall be selected in accordance with inspection level I and the AQL shall be 2.5 percent defective.

4.3.2 Examination procedure. An examination shall be made to determine the compliance of the packaging, packing, and marking with the requirements of section 5. The sample units shall be examined for the defects described below and scored accordingly.

Examination of Preparation for Delivery

Examine	Defects
Packaging	Not as specified (see 5.1).
Packing	Not as specified (see 5.2).
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Contents	Presence of lumps or caking.

4.4 Testing.

4.4.1 Sampling for testing. The lot size for testing shall be defined as the number of filled containers that contain the material manufactured within a 24-hour period in a single operation at one location using the same process and raw materials. The sample unit shall be defined as one such filled container. A 1-pound composite shall be made by mixing equal portions of each of the sample units taken from the lot in accordance with inspection level S-1. The lot shall be unacceptable if the composite sample fails to pass any of the tests described in 4.4.2.

4.4.2 Testing procedures.

4.4.2.1 Fumarin determination. Weigh an amount of reagent fumarin consisting of 300.0 mg pure fumarin into a 100-ml volumetric flask and fill to volume with 2 molar aqueous NaOH. A 4.0-ml aliquot of the resulting solution is transferred to a 1,000-ml volumetric flask and diluted to volume with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 304 and 306 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

Weigh 10.00 gm of the rodenticide into a 100-ml volumetric flask and fill to volume with dilute aqueous NaOH. Transfer a 40-ml aliquot to a 1,000-ml volumetric flask and dilute to volume with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 304 and 306 nm into a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

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The percentage of fumarin in the rodenticide is calculated by using the following formula:

$$\text{Percent} = 10\left[\frac{A_2}{A_1} \times \frac{W_1}{W_2}\right]$$

where: A_1 = absorbance of pure fumarin solution.
A_2 = absorbance of rodenticide sample solution.
W_1 = milligrams of pure fumarin used.
W_2 = grams of rodenticide sample used.

4.4.2.2 Pival determination. Weigh an amount of reagent pival consisting of 300.0 mg pure pival into a 100-ml volumetric flask and fill to volume with 2 molar aqueous NaOH. A 4.0-ml aliquot of the resulting solution is transferred to a 1,000-ml volumetric flask and diluted to volume with dilute NaOH. Pipette a sufficient amount of this solution to a 1-cm quartz cuvette and determine its maximum absorbance found between 283 and 286 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

Weigh 10.00 gm of the rodenticide into a 100-ml volumetric flask and fill to volume with dilute aqueous NaOH. Transfer a 40-ml aliquot to a 1,000-ml volumetric flask and dilute to volume with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 283 and 286 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

The percent of pival in the rodenticide is calculated by using the following formula:

$$\text{Percent} = 10\left[\frac{A_2}{A_1} \times \frac{W_1}{W_2}\right]$$

where: A_1 = absorbance of pure pival solution.
A_2 = absorbance of rodenticide solution.
W_1 = milligrams of pure pival used.
W_2 = grams of rodenticide sample used.

4.4.2.3 Diphacinone determination. Weigh an amount of reagent diphacinone consisting of 100.0 mg pure diphacinone into a 100-ml volumetric flask and fill to volume with 2 molar aqueous NaOH. A 10.0-ml aliquot of the resulting solution is transferred to a 1,000-ml volumetric flask and diluted to volume with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 286 and 288 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

Weigh 10.0 gm of the rodenticide into a 100-ml volumetric flask and fill to volume with dilute aqueous NaOH. Dilute a 20-ml aliquot to 100 ml with dilute NaOH in another 100-ml volumetric flask. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 286 and 288 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a denatured alcohol blank.

The percent of diphacinone in the rodenticide is calculated by the following formula:

$$\text{Percent} = 10\left[\frac{A_2}{A_1} \times \frac{W_1}{W_2}\right]$$

where: A_1 = absorbance of pure diphacinone solution.
A_2 = absorbance of rodenticide solution.
W_1 = milligrams of pure diphacinone used.
W_2 = grams of rodenticide sample used.

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4.4.2.4 Chlorophacinone. Weigh an amount of reagent chlorophacinone consisting of 100.0 mg pure chlorophacinone into a 100-ml volumetric flask and fill to volume with 2 molar aqueous NaOH. A 10-ml aliquot of the resulting solution is transferred to a 1,000-ml volumetric flask and diluted to volume with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 324 and 326 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

Weigh 10.00 mg of the rodenticide into a 100-ml volumetric flask and fill to volume with dilute aqueous NaOH. Dilute a 20-ml aliquot to 100 ml with dilute NaOH in another 100-ml volumetric flask. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 324 and 326 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

The percentage of chlorophacinone in the rodenticide is calculated by using the following formula:

$$\text{Percent} = 10\left[\frac{A_2}{A_1} \times \frac{W_1}{W_2}\right]$$

where: A_1 = absorbance of pure chlorophacinone solution.
A_2 = absorbance of rodenticide solution.
W_1 = milligrams of pure chlorophacinone used.
W_2 = grams of rodenticide sample used.

4.4.2.5 PMP determination. Weigh an amount of reagent PMP consisting of 100.0 mg pure PMP into a 100-ml volumetric flask and fill to volume with 2 molar aqueous NaOH. A 10.0-ml aliquot of the resulting solution is transferred to a 1,000-ml volumetric flask and diluted to volume with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 282 and 284 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

Weigh 10.0 gm of the rodenticide into a 100-ml volumetric flask and fill to volume with dilute aqueous NaOH. Dilute a 20-ml aliquot to volume in a 1,000-ml volumetric flask with dilute NaOH. Pipette a sufficient amount of this solution into a 1-cm quartz cuvette and determine its maximum absorbance found between 282 and 284 nm in a Beckman spectrophotometer model D.U., or its equivalent, using a dilute NaOH blank.

The percent of PMP in the rodenticide is calculated by using the following formula:

$$\text{Percent} = 10\left[\frac{A_2}{A_1} \times \frac{W_1}{W_2}\right]$$

where: A_1 = absorbance of pure PMP solution.
A_2 = absorbance of rodenticide solution.
W_1 = milligrams of pure PMP used.
W_2 = grams of rodenticide sample used.

4.4.2.6 Tetrasodium ethylenediamine tetraacetate (EDTA) determination. Prepare the following solutions: 500 ml of 0.20 M Na_2HPO_4 with pH adjusted to 11.0 with NaOH, 4.0 gm EDTA in 1,000 ml H_2O, 0.25 percent $CuSO_4$ in H_2O. Dilute 3.0 ml, 3.5 ml, and 4.0 ml aliquots of the EDTA solution to volume in three 100-ml volumetric flasks with the Na_2HPO_4 buffer. Dilute a 10-ml aliquot of the rodenticide solution to volume in a 100-ml volumetric flask with the Na_2HPO_4 buffer. Titrate the four buffered solutions with $CuSO_4$ solution until the first permanent turbidity and then add two drops excess. Determine the absorbancies of the four titrated solutions at 340 nm using a Beckman model D.U. quartz spectrophotometer, or equivalent, and a blank with Na_2HPO_4 solution. Plot the absorbancies of the three standard EDTA solutions against the corresponding EDTA concentrations in the buffered standards.

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The percent of EDTA in the rodenticide is calculated from the concentration in the buffered solution, determined with the standard curve, using the following formula:

$$\text{Percent} = \frac{(\text{concentration})\ (10{,}000)}{4.5}$$

4.4.2.7 Sodium benzoate determination. Determine the sodium benzoate in the sample using the method for sodium benzoate outlined in the United States Pharmacopeia.

4.4.2.8 Particle size. Transfer 200 grams of the rodenticide to a 60-mesh sieve, conforming to RR-S-366, and screen in a Ro-tap, or equivalent machine, for 10 minutes. Stop the machine. Break up any soft lumps of caked material by brushing lightly over the sieve, and restart the machine. Completion of screening is determined by shaking the sieve in the hand over a hard surface and noting if any noticeable amount passes the screen. Weigh residue remaining on the screen. Calculate the portion passing through the 60-mesh screen, and express in percent.

4.4.2.9 Moisture. Weigh approximately 2 grams to the milligram in a tared moisture dish and dry for 10 hours in a drying oven at a temperature of 100° ± 2°C; cool in a desiccator and reweigh. The loss due to moisture shall be reported as the percent of the sample weight.

4.4.2.10 Performance. The performance test shall use bio-assay animals to determine the suitability of the rodenticide by determining the acceptability and effectiveness of the rodenticide concentrate used in two test baits. The bio-assay animals shall consist of mature, healthy, male albino rats of a Wistar strain weighing between 150 and 250 grams.

4.4.2.10.1 Liquid bait test. The animals used in this test shall have been maintained on a control diet consisting of pelleted laboratory chow, dog biscuits, or equivalent laboratory animal food. Twenty-four hours prior to the start of the test period, 20 mature, healthy, male albino rats are to be transferred to individual feeding cages supplied with water and the control diet.

The liquid bait used in this test shall be made by dissolving 25.0 grams of the rodenticide in sufficient distilled water to make 946 ml of solution. The control diet and the test solution are to be supplied in excess for a period of 8 consecutive days. No liquids other than the test solution shall be available during the test period. At the end of the 8-day assay period, the test solution is to be removed, and the animals given control diet and water for an additional observation period of 10 days. The temperature of the room shall be kept between 65° and 85°F (18° and 27°C) throughout the test period.

4.4.2.10.2 Cereal bait test. Thirty mature, healthy, male albino rats are to be deprived of all food for a period of 24 hours, then placed in individual feeding cages. Mix 850 grams of dried bread crumbs with 100 grams of rodenticide. Add 50 grams of petrolatum liquid and mix thoroughly by blending with a spatula on a sheet of glazed paper, followed by repeated screening through a 20-mesh screen.

The cereal bait and water are to be supplied in excess for a period of 8 consecutive days. At the end of the period, the test ration is to be removed, and the animals given control diet and water for an additional observation period of 10 days. The temperature of the room shall be kept between 65° and 80°F (18° and 27°C) throughout the test period.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. One pound of rodenticide concentrate shall be packaged in a round, multiple friction top can conforming to PPP-C-96, type V, class 2, plan B coating.

5.1.2 Level C. Rodenticide concentrate shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Rodenticide concentrate shall be packed in accordance with the appendix to PPP-C-96 for level A packing.

5.2.2 Level B. Rodenticide concentrate shall be packed in accordance with the appendix to PPP-C-96 for level B packing.

5.2.3 Level C. Rodenticide concentrate shall be packed in a container complying with 49 CFR 170-178 to assure carrier acceptance and safe arrival at destination.

5.3 Marking. Marking of the unit and shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies.

6. NOTES

6.1 Intended use. The rodenticide concentrate is intended for use in conjunction with a suitable liquid or cereal bait for the control of rats and mice.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packing and packaging (see 5.1 and 5.2).
(c) Applicable marking document (see 5.3).

6.3 Anticoagulant names. The chemical names analogous to the generic terms given in table I are as follows:

Pival is 2-pivalyl-1,3-indandione.
Fumarin is 3-(alpha-acetonylfurfuryl)-4-hydroxycoumarin.
Diphacinone is 2-diphenylacetyl-1,3-indandione.
Chlorophacinone is 2-((p-chlorophenyl)phenylacetyl)-1,3-indandione.
PMP is 2-isovaleryl-1,3-indandione.

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